

The project looks at how an immersive VR environment could address these challenges.

The team led by Prof. Sylvie Delacroix, professor of law and ethics, University of Birmingham, sets out that VR has a very important role to play in helping GPs further develop the wide range of intuitive and perceptual skills that are essential to their practice.

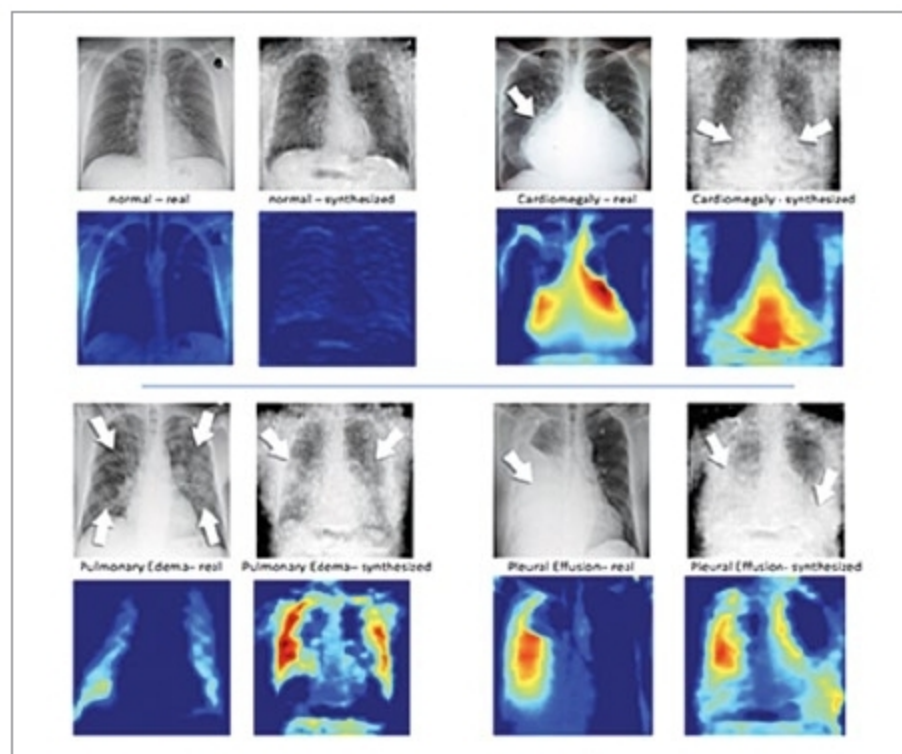
Dr Sylvia Xueni Pan, Goldsmiths, University of London, led the development of an immersive VR environment replicating a GP's surgery, where GP participants were able to interact with virtual, avatar patients in the same way as they would in a real consultation.

In the virtual scenario, the GP encountered a patient with a complex medical condition. He was accompanied by his son, whom he treated so aggressively as to cause the son to flinch. The tell-tale signals of child abuse were made subtler in one condition, and more obvious in the other, with expert advice provided by Dr Caroline Fertleman, University College London, a paediatrician with expertise in child-safeguarding. All characters were animated using motion capture data captured from professional actors. These animation and dialogue elements were then integrated into the animation engine.

A total of 64 GPs with different levels of experience were involved in the study. Its aim was to establish whether levels of experience would have any impact on the GPs' ability to pick up child-safeguarding concerns, and whether more experienced GPs would find it easier to pick up subtle, rather than obvious, signs of child-safeguarding concerns.

Creating computer-generated X-rays to augment AI training sets

Identifying rare pathologies in medical images has been a persistent challenge for researchers, because of the scarcity of images that can be used to train artificial intelligence (AI)



On the left of each quadrant is a real X-ray image of a patient's chest and besides it, the synthesized X-ray formulated by DCGAN. Under X-ray images are corresponding heatmaps, which is how the machine learning system sees the images (Credit: Machine Intelligence in Medicine Lab)

systems in a supervised learning setting. Prof. Shahrokh Valaee from The Edward S. Rogers Sr. Department of Electrical & Computer Engineering (ECE) and his team have designed a new approach that uses machine learning to create computer-generated X-rays to augment AI training sets.

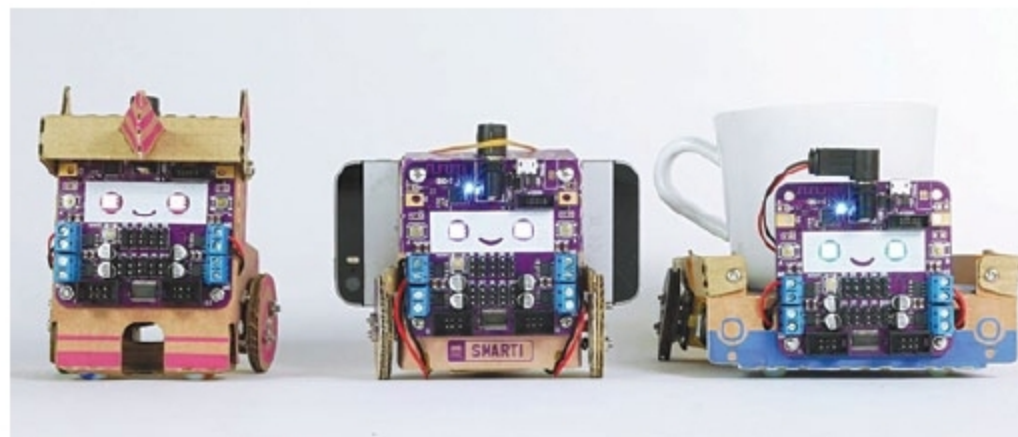
To create these artificial X-rays, the team uses an AI technique called deep convolutional generative adversarial network (DCGAN) to generate and continually improve simulated images. GANs are a type of algorithm made up of two networks: one that generates images and the other that tries to discriminate synthetic images from real images.

The two networks are trained to the point that the discriminator cannot differentiate real images from synthesized ones. Once a sufficient number of artificial X-rays are created, these are combined with real X-rays to train a deep convolutional neural network, which then classifies the images as either normal or identifies a number of conditions.

Prof. Valaee is a member of Machine Intelligence in Medicine Lab (MIMLab)—a group of physicians, scientists and engineering researchers who are combining their expertise in image processing, AI and medicine to solve medical challenges.

World's first AI-enabled cardboard robot

Smartibot is the world's first AI-enabled cardboard robot that you can build yourself. It works with a smartphone, allowing you to use your cellphone as a remote control, or as its brain by attaching it to your robot.



Create a number of robots using the circuit board, motors and battery box (Credit: www.kickstarter.com)

Smartibot app contains YOLO, a powerful AI, which can recognise objects Smartibot sees, including people, cars, dogs and bicycles, among others. Without any programming, you can tell your robot to follow you around with a cup of tea, chase your cat out of your bedroom or follow your toy car.

You can connect up to 14 motors (four DC and 10 servos) to build wheeled robots, two-legged robots, four-legged robots, robot arms, hovercrafts and much more. The app allows you to design custom controllers to suit your robot and define how it behaves without needing to write any code.

Apart from AI robot, Smartibot comes with cardboard parts to make two additional robots. With Smartibot you can build robots out of almost anything around the house. Re-use the circuit board, motors and battery box, and attach these to your creations.